An independent exact recheck of the PG(2,9) multiple-unicast counterexample

(a recheck of a result by others; no novelty is claimed)

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AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, ACCEPT_WITH_CORRECTIONS), not by a human.** The corrections are applied (Section 8). No human audit exists. There is no Lean in this paper.

Abstract

M. Braverman and Z. He [1] (arXiv:2610.10108, 7 Oct 2026, unrefereed) give a network on 182 vertices, obtained from the incidence graph of the projective plane $PG(2,9)$ by deleting 175 edges, with 157 unicast sessions, and a deterministic causal linear code over $\mathbb{F}_9$ (their Section 4) that, they say, beats the best fractional multicommodity flow, and so refutes the Li–Li / Harvey–Kleinberg–Lehman conjecture for undirected networks. The paper says GPT-6 found the non-deterministic example of its Section 3; **the claim checked here is the deterministic Section 4 code. The result is theirs; we claim nothing new.** With our own code, and an auditor's independent code (Sage), we reproduce the finite facts exactly: the graph (735 edges, every session at distance exactly 5, total 785), the $\mathbb{F}_9$ arithmetic, all 182 local codes 5-dimensional MDS, the rank 744 of the 910×910 parity matrix ($\dim W = 166$, $\dim W_C = 157$, injective on the session edges), the Appendix B assignments ($\dim W = 166$ exactly), decoding of all 157 unit inputs (hence, by linearity, of every input), and the exact optimum 147/157 of the fractional flow (an exact rational flow plus the length-function dual). Not verified: that the paper's model is the conjecture's model, the amplification theorem, Section 3, the authors' verifier. We do not say the conjecture is resolved beyond what is listed. Outcome: confirmed as a recheck; Tier A (Lean) not reached.

What this is NOT. Not a new result; not a verification that the conjecture, as stated by Li–Li and Harvey–Kleinberg–Lehman, is refuted (the model equivalence is Tier [L], not checked); not a check of the amplification (Corollary 1.2) or of Section 3; not a run of the authors' verifier; not a formal proof (no Lean); not human-audited; not an exhaustive prior-art search.

Markers. [L]: a cited source, taken as stated. [C]: executed exact code (finite-field, integer and rational arithmetic) or an elementary argument written down, not formalised. [X]: random-input simulation and floating-point readings, and statements of what was not done. Identifiers `MUC-··-nnnn` refer to `ledger/ledger.json`.

1 The claim and its scope

The conjecture (Li–Li 2004; Harvey–Kleinberg–Lehman 2004; [L]) says that in undirected networks coding gives no throughput advantage over fractional multicommodity flow [1]. Braverman and He take H, the point–line incidence graph of $PG(2,9)$: 182 vertices, 10-regular, 910 edges, bipartite, girth 6. They delete a set R of 175 edges and keep $G = H - R$ with 735 edges. Sets $C, J \subset R$ of 9 edges each give $B = R \setminus (C \cup J)$, $|B| = 157$, one unicast session per edge of B (the earlier endpoint in a vertex order π is the source, the later

one the sink). Physical edges are oriented earlier→later in π, deleted ones later→earlier. The code lives over $\mathbb{F}_9 = \mathbb{F}_3[i]/(i^2+1)$: at each vertex the incident values, indexed by the neighbours, lie in a 5-dimensional local code (evaluations of a binary quartic form), so any 5 values determine the other 5. The claim is that every sink recovers its source symbol while all sources have rate 1 symbol, i.e. 735 symbol transmissions serve sessions whose total distance is $785 > 735$, so routing is at most $735/785 = 147/157 < 1$. Section 3 of [1] contains a non-deterministic variant (166 deleted edges, 744 edges, bound 744/830); the paper credits GPT-6 with it. It was not recomputed here (`MUC-L-0001`). The lists R, π, C, J of its Appendix A are third-party content and are not redistributed (Section 10).

2 Graph and sessions ([C])

Both implementations build $PG(2,9)$ from the 91 normalised triples in the paper's order. Results: 182 vertices, 910 edges, 10-regular, bipartite, girth 6 (any two points share exactly one line); $|R| = 175$ distinct indices, C, J disjoint 9-subsets of R, $|B| = 157$, π a permutation of $0, \ldots, 181$; G has 735 edges and is connected; on the full 910-edge orientation $d^-(v) = d^+(v) = 5$ and $d^+_C(v) = d^-_J(v)$ at all 182 vertices. There are 157 sessions from 66 distinct sources, and the BFS distance in G of every session is exactly 5 (sum 785). Also $735 < 785$ and $735/785 = 147/157$ in rational arithmetic (`MUC-C-0001`).

3 Field, local codes and rank ([C])

Field. The producer's hand-built tables for $\mathbb{F}_9$ satisfy associativity, distributivity, commutativity, inverses, no zero divisors, cyclic multiplicative order 8 and characteristic 3; the auditor used Sage `GF(9, modulus=x^2+1)` and cross-checked with 2×2 blocks over $\mathbb{F}_3$ (a ring homomorphism on all of $\mathbb{F}_9$). *Local codes.* At all 182 vertices the local code has dimension 5 and is MDS: all 252 five-subsets of the 10 incident positions are independent (`MUC-C-0002`).

Global rank. The 910×910 parity matrix has rank 744 over $\mathbb{F}_9$, so $\dim W = 166$; the $\mathbb{F}_3$ block version has rank $1488 = 2 \cdot 744$. With the public zeros on C, $\dim W_C = 157$, and $\operatorname{rank}[P; e_C; e_B] = 910$, so restriction of W_C to the session edges B is injective: $W_C \cong \mathbb{F}_9^{157}$ (`MUC-C-0003`).

Appendix B of [1]. The auditor checked that the 166 retained bidegree-$(4,4)$ monomials ($225 - 30 - 30 + 1 = 166$) give assignments of rank 166 that satisfy every local code, and that all 225 monomials together span only rank 166. Hence $\dim W = 166$ exactly: the paper's lower bound is tight on this instance, and W is the span of the Appendix B assignments. The paper's analytic proof was not re-derived (`MUC-C-0004`).

4 Causal decoding ([C], [X])

Vertices are processed in the order π. At each vertex there are exactly 5 known independent values (incoming physical edges, own B-sources, $C = 0$); the vertex interpolates, emits the other 5, and the checks are consistency on incoming edges, agreement at both ends of J-edges, and decoding at each sink. The auditor's run: 157/157 **unit-vector inputs decode correctly**. The whole map is $\mathbb{F}_9$-linear in the input, so the unit vectors cover all 9^{157} inputs; this exact argument is Tier [C] (audit correction 3). Random inputs, producer 1000/1000 and auditor 200/200, are a redundant [X] check (`MUC-C-0005`, `MUC-X-0001`).

5 The flow optimum is exactly 147/157 ([C])

Upper bound. The length function $\ell_e = 1$ gives load $\geq r \cdot 785$ against total capacity 735, hence $r \leq 147/157$. *Exact primal.* The producer's HiGHS value 0.9363057324840819 against $147/157 = 0.9363057324840764$ was only a floating-point agreement. The auditor built an exact flow: a feasibility LP over shortest-path-DAG arcs (66 source groups, 31,899 arc variables) with each session demand 147/157 and each edge load 1; HiGHS returned a vertex with support 3,328 columns of full column rank; that support system was solved exactly over $\mathbb{Q}$ in Sage (largest denominator about $1.7 \cdot 10^{101}$), then re-verified in Python `Fraction`: flows nonnegative, exact conservation with demand 147/157 for every session, every edge load 1. The run was repeated, with the same support and denominators. The certificate itself (3,328 arc flows, sha256 `5f29f1ac...9ed3`, 715,658

bytes) is *withheld*: the complement of its edge support determines the third-party deleted-edge set R. We ship only its hash and a summary (`data/audit/outputs/lp_certificate_summary.json`); it is regenerated by `data/audit/code/lp_exact.py` from the locally extracted lists (see `data/THIRDPARTY_FETCH.txt`), and the reader can compare the hash. So the undirected unit-capacity max concurrent flow optimum is *exactly* 147/157, and every session uses only shortest paths and every edge is saturated (`MUC-C-0006`).

6 Rate model ([C] argument, [L] for the model)

Under the paper's Section 2 definition (each edge split into two directed edges, $H(X_{\rightarrow}) + H(X_{\leftarrow}) \leq c(e)c^*$, throughput r with $H(m_i) \geq r\, d_i\, c^*$; here $c^* = \log 9$), *one* execution already achieves $r = 1$: uniform $x_q \in \mathbb{F}_9$ gives $H(m_i) = \log 9$, and each physical edge carries one symbol in one direction only. Causality and correctness are exactly what Section 4 checks. Pipelining is needed only for the time-slot wording of Section 4 of [1]. That this Section 2 model is the model of the conjecture of Li–Li and Harvey–Kleinberg–Lehman is Tier [L] and was not checked (`MUC-C-0007`, `MUC-L-0003`).

7 Controls

Run before the headline (`MUC-C-0008`). Positive: the directed butterfly, where the GF(2) XOR code decodes all 4 inputs at rate 1 while routing gets 0.5; the undirected 4-cycle with two opposite sessions (LP 1.0), a single edge (1.0) and a path with two sessions (0.5); the true field passes every check. Negative: the field relations $t^2 = 0$ and $t^2 = 1$ fail; a tampered R (restore edge 7, delete edge 0) breaks $d^- = d^+ = 5$ at 2 vertices and 20/20 causal trials fail; renormalising (λ, μ) at all neighbours of one vertex (the paper says not to) fails 20/20; adding 1 to one transmitted symbol fails 20/20; scaling one local-code row by θ fails 17/20 in the producer's run and 16/20 in the auditor's (the others pass by chance, plausibly because the affected value is 0; not checked edge by edge). An earlier renormalisation control that touched only one neighbour passed and was replaced, since it was a flaw in the control.

8 Audit corrections applied

The producer's `data/RESULT.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them on the audit's eight corrections. (1) The flow optimum is certified exactly, not only floating. (2) Item A11 is exact: an exact rational flow of value 147/157 with every edge saturated, plus the dual $\ell_e = 1$. (3) Tiers: [C] for decoding via the 157 unit inputs plus linearity and for A11; the random trials are a redundant [X] check; [L]: that the Section 2 model is the LL04/HKL04 model, the BGS17 amplification and Corollary 1.2. (4) The plan item reads: GPT-6 found the non-deterministic (Section 3) example; the checked claim is Section 4. (5) The Appendix A lists were re-extracted by the auditor with two tools (`pdftotext -layout` and PyMuPDF word boxes), which agree element by element after dropping the page-number footer "13"; hashes are identical to the producer's transcription. (6) Appendix B is verified as in Section 3. (7) For [2], the auditor read the abstract: a completely different counterexample (below). (8) The preregistration (`PREREG_ITEM.md`, sha256 `a0582465...eb29`) was committed in the same commit as `RESULT.md` (a06dcfd); that it came first is shown only by file timestamps (07:17 against first code at 07:18).

9 Prior art

A same-day, independent counterexample exists: Zhang, Li and Li [2] (arXiv:2610.09367, 7 Oct 2026) use a binary linear block code on a connected network of maximum degree 3 with leaf terminals, built from a reversible circuit, with their own tensor amplification; they state that their finite counterexample and family theorem are formalised in Lean, which was *not checked* here. Follow-ups 2610.20079 and 2610.33655 and attempts 2608.06042 and 2608.06070 were found but not read. The OpenAI preprint "Finite tensor savings and exact Fourier circuits", which [1] cites, was not found or opened. The kingy.ai and MathWorld statements quoted in `RESULT.md` are unverified. No independent replication of the authors' verifier was found. This recheck therefore adds only one more independent computation of one of two same-day counterexamples.

10 Limitations and what is not done

- AI audit only; no human audit; no Lean (no Tier A).
- Unverified (Tier [L]): the Section 2 model equivalence with the conjecture, BGS17 amplification and Corollary 1.2, Section 3's example (166/744/830), the authors' verifier and repository (not run), the OpenAI preprint, the Lean formalisation claimed by [2], the kingy.ai/MathWorld statements.
- The statement "refutes the conjecture" therefore rests on the model equivalence and is not asserted here beyond the verified finite facts.
- Third-party content: the Appendix A lists R, π, C, J and the producer's transcription file are not shipped, nor the derived session-pair list; `data/THIRDPARTY_FETCH.txt` gives their hashes and extraction instructions, so the scripts do not run without them. The exact LP certificate is also withheld (hash and summary only), since it determines R.
- Trusted components: numpy, Sage, scipy/HiGHS, PyMuPDF, pdftotext.

11 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Producer code, data and logs are in `data/code`, `data/data`, `data/logs`; the auditor's scripts (`extract.py`, `audit.py`, `appB.py`, `lp_exact.py`) and small outputs are in `data/audit`. The third-party full texts are not shipped. Nothing was committed, pushed or uploaded.

References

[1] M. Braverman, Z. He, *The Multiple Unicast Conjecture is False*, arXiv:2610.10108 (v1, 7 Oct 2026), unrefereed. ([L])

[2] X. Zhang, B. Li, Z. Li, *Network Coding Can Beat Routing in Undirected Multiple-Unicast Networks*, arXiv:2610.09367 (v1, 7 Oct 2026), unrefereed. ([L])